

ASIAN HANDICAP MARKET MAKING: END-TO-END SYSTEM FLOW

Statistical Signal Discovery → Bayesian Valuation → Guéant-Cartea Pricing → Asymmetric Quoting → Active Auto-Hedge

STEP 01 SIGNAL DISCOVERY

- **Poisson Forecasting:**
Dixon-Coles bivariate goal intensity
- **Sharp Market Feed:**
Pinnacle/Macao consensus & de-vigging
- **Volatility Metric:**
Real-time standard deviation σ_t

P_{fore}, P^*

$$P_{\text{fore}} = \mathbb{P}(\Delta G + H > 0) + \frac{1}{2}\mathbb{P}(\Delta G + H = 0)$$

$$P_{\text{sharp}}^* = \frac{q_H}{q_H + q_A}, \quad \sigma_t = \text{Std}(P^*)$$

STEP 02 VALUATION & PRICING

- **Bayesian Blending:**
Time-decay weighting: $w(\tau) \rightarrow 1$ as $\tau \rightarrow 0$
- **Terminal Penalty:**
Guéant inventory risk $U(\tau) = 1/\sqrt{\tau + \epsilon}$
- **Alpha Drift:**
Cartea short-term momentum $\theta_\alpha \propto \tau$

$P_{\text{mid}}(t)$

$$P_{\text{fair}}(t) = w(\tau)P_{\text{sharp}}^* + [1 - w(\tau)]P_{\text{fore}}$$

$$P_{\text{mid}}(t) = P_{\text{fair}}(t) - \lambda I_t^3 U(\tau) + \theta_\alpha \alpha_t^\tau$$

STEP 03 ASYMMETRIC QUOTING

- **Inventory Skew:**
Widen spread against accumulated risk I_t
- **Volatility Shift:**
Widen spread during volatile intervals
- **Published Quotes:**
Continuous two-way limit orders [Bid, Ask]

Quotes

$$\delta_t^b = \delta_0 + \gamma I_t + \psi \sigma_t, \quad \delta_t^a = \delta_0 - \gamma I_t + \psi \sigma_t$$

$$\text{Bid} = P_{\text{mid}} - \delta_t^b, \quad \text{Ask} = P_{\text{mid}} + \delta_t^a$$

STEP 04 EXECUTION & DEFENSE

- **Wholesale Fills:**
Syndicate limit order fills: $\lambda(\delta) = A e^{-k\delta}$
- **Circuit Breaker:**
Halt quotes if order toxicity $\kappa_t > 0.20$
- **Auto-Hedge Sweep:**
Aggressive de-risking if $|I_t| > 15\% I_{\text{max}}$

$$\text{Fill Rate: } \lambda(\delta) = A e^{-k\delta} \quad (\text{Ticket } 1k - 3k\$)$$

$$\text{If } |I_t| > 0.15 I_{\text{max}} \Rightarrow \text{Sweep at } \leq 50 \text{ bps}$$

CLOSED-LOOP DEFENSE

Real-Time Inventory Feedback & Auto-Hedge De-risking Channel

Inventory Position I_t Feedback

Fill Events $\Rightarrow \Delta I_t$